You'll enable TLS for requests coming into the network from external clients in section 10.4.

Mutual TLS

Linkerd and most other service meshes don't just supply normal TLS server certificates, but also client certificates that are used to authenticate the client to the server. When both sides of a connection authenticate using certificates this is known as *mutual TLS*, or *mutually authenticated TLS*, often abbreviated *mTLS*. It's important to know that mTLS is not by itself any more secure than normal TLS. There are no attacks against TLS at the transport layer that are prevented by using mTLS. The purpose of a server certificate is to prevent the client connecting to a fake server, and it does this by authenticating the hostname of the server. If you recall the discussion of authentication in chapter 3, the server is claiming to be `api.example.com` and the server certificate authenticates this claim. Because the server does not initiate connections to the client, it does not need to authenticate anything for the connection to be secure.

The value of mTLS comes from the ability to use the strongly authenticated client identity communicated by the client certificate to enforce API authorization policies at the server. Client certificate authenticate is significantly more secure than many other authentication mechanisms but is complex to configure and maintain. By handling this for you, a service mesh enables strong API authentication mechanisms. In chapter 11 you'll learn how to combine mTLS with OAuth2 to combine strong client authentication with token-based authorization.

The current version of Linkerd can automatically upgrade only HTTP traffic to use TLS, because it relies on reading the HTTP Host header to determine the target service. For other protocols, such as the protocol used by the H2 database, you'd need to manually set up TLS certificates.

> TIP Some service meshes, such as Istio, can automatically apply TLS to non-HTTP traffic too.[5] This is planned for the 2.7 release of Linkerd. See *Istio in Action* by Christian E. Posta (Manning, 2020) if you